

Pencil Code User Meeting 2026

PCUM-26 · 21–25 September 2026 · University of Alicante

All times local (CEST, UTC+2)

	Monday 21 <i>getting started & GPUs</i> <i>collecting coding topics</i>		Tuesday 22 <i>community & development</i>	Wednesday 23 <i>chiral MHD & dynamos</i>	Thursday 24 <i>numerical dissipation & PINN</i>	Friday 25 <i>final talks & wrap-up</i>
09:30 – 10:15	Introductions 2 min per person		Steering Committee Report	Deepen Garg	Simon Candelaresi	Alberto Roper Pol
10:15 – 11:00	Axel Brandenburg		Philippe Bourdin	Simon Candelaresi	Jorge Urban	Ramkishor Sharma <i>online</i>
11:00 – 11:30	Coffee		Coffee + group photo	Coffee	Coffee	Coffee
11:30 – 13:00	Tutorial: beginners Touko Puro	Tutorial: admin Philippe Bourdin	<i>Coding</i> Axel & Touko	<i>Coding</i> Storing large arrays	<i>Coding</i> Finding pencils	Brainstorming GR in the Pencil Code Matthias Rheinhardt
13:00 – 15:00	LUNCH		LUNCH	LUNCH	LUNCH	FAREWELL LUNCH
15:00 – 15:45	Tutorial: GPU Touko Puro		EXCURSION	Clara Dehman	Vinay Kumar <i>online</i>	Concluding remarks + next year's agenda
15:45 – 16:30	<i>Coding</i> GPU			<i>Coding</i> Symplectic solver	<i>Coding</i> Numerical dissipation	FREE TIME
16:30 – 17:00	Coffee			Coffee	Coffee	
17:00 – 18:00	Neco Kriel			<i>Coding</i> Subcycling	<i>Coding</i> Numerical dissipation	
<i>evening</i>	Welcome drinks informal				Conference dinner	

Talks

Axel Brandenburg – Pencil Code since 2025; admin tutorial (Mon)

Philippe Bourdin – Recent topics in Pencil Code development

Alberto Roper Pol – *Relativistic MHD in an expanding Universe with the Pencil Code*

Touko Puro – tutorials: beginners (Mon am), GPU (Mon pm)

Deepen Garg – Chiral MHD in Pencil Code

Clara Dehman – Neutron star physics

Ramkishor Sharma (online) – Magnetogenesis from a pseudoscalar–gauge field interaction

Jorge Urban – Hydrodynamics with physics-informed neural networks

Neco Kriel – the Pencil and Quokka codes for small-scale dynamos

Simon Candelaresi – Numerical dissipation in CFD codes (Wed). Multiphysics coupling (Thu).

Vinay Kumar (online) – Does the plasmoid-mediated reconnection rate depend on the magnetic Prandtl number?

Coding topics

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Collected on Monday and worked on in the collaborative coding sessions through the week

- Python and IDL routines
- Spectral output
- Persistent variables
- GPU
- Symplectic solver
- Subcycling
- Finding pencils
- Numerical dissipation
- Magic and missing pencils